

Algorithmic Pseudocode Sample 35

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Require: trajectory τ

```
1: for layer  $i$  in HKSL (begin with the lowest layer) do
2:   for layer  $j$  in HKSL (begin with the highest layer) do
3:     if  $i == j$  then
4:       break loop
5:     end if
6:     Embed first observation  $o$  in  $\tau$  using layer  $j$ 's encoder
7:     if layer  $j$  is the top layer then
8:       for step layer  $j$  can take in  $\tau$  do
9:         Compute forward-step using layer  $j$ 's forward model and actions from  $\tau$ 
10:        Store the prediction
11:      end for
12:    else layer  $j$  is not the top layer
13:      for step layer  $j$  can take in  $\tau$  do
14:        Compute forward-step using layer  $j$ 's forward model, actions from  $\tau$ , and output
from communication manager using the stored rollout from above level
15:      Store the prediction
16:    end for
17:  end if
18: end for
19: Embed first observation  $o$  in  $\tau$  using layer  $i$ 's encoder
20: for step layer  $i$  can take in  $\tau$  do
21:   Compute forward-step using layer  $i$ 's forward model, actions from  $\tau$ , and output from
communication manager using the stored rollout from above level
22:   Project the forward model's output with layer  $i$ 's nonlinear projection
23:   Compute loss per Equation ??
24: end for
25: Update layer  $i$ 's weights
26: end for
```
